Exercise 5

NULL function

Question 1

Show all employees with their Salary, if Salary is NULL, display 0

Syntax

```
Select Employee_ID,
       Name,
   IfNull(Salary, 0) AS Salary_with_default
From Employees;
```

Employee_ID	Name	Salary_with_Default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Question 2

Show employee names with their department, if department is Null, show "Not Assigned"

Syntax

```
Select Employee_ID,
       Name,
   IfNull(department, 'Not Assigned') AS department_name
From Employee;
```

Employee_ID	NAME ~~DANA~~	Department_Name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	DANA	Finance

Question 3

Find orders with NULL customer_ID using ISNULL()

Syntax

```
SELECT order_ID,
       customer_ID)
  from Orders
  WHERE ISNULL (customer_ID);
```

Order_ID	customer_ID
103	NULL

Question 4

Show all orders. If delivery_date is Null, Show 'Pending'

Syntax

```
SEleet order_id,
       Customer_id,
  IFNull (delivery_date, 'Pending') As delivery_Status
from Orders;
```

order_id	Customer_ID	delivery Status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Question 5

Show all Students and their grades. Replace Null with 0

Syntax

```
Select Student_id
       name,
       IfNull (grade, 0) As final_grade
from Students;
```

Student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Question 6

Count Students who haven't been graded

Output

Not_graded_count
1

Syntax

~~Select Count~~

```
Select Count (*) As not_graded_count
  from Students
  where grade is Null;
```

Question 7

Show product name, Price and finalPrice after discount (assume 0 if discount is Null)

```
Select Product_id,
       Name,
       Price - IfNull (discount, 0) As Final_Price
from Products;
```

Product_id	Name	Final_Price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

Question 8

Count how many customers have no email.

Syntax

```
select count(*) As Missing_email_count
  from customers
  where email is Null;
```

Missing_email_count
2

Question 9

Show all customers with email, if Null, display "NoEmail"

Syntax

```
select customer_id
       name,
    IfNull (email, 'NoEmail') AS email_display
  from customers;
```

customer_id	name	email_display
1	linda	NoEmail
2	Joseph	Joseph@email.com
3	Nia	NoEmail

Question 10

Show Payment details with method replaced by "Unknown" if Null

Syntax

```
Select Payment_id,
       IfNull (Method, 'Unknown') As method_display,
       Status
from Payments;
```

Output

Payment_id	Method_display	Status
301	Credit	Null
302	PayPal	Success
303	Unknown	Failed

Question 11

Show items and their quantity (0 IFNull)

Syntax

```
Select item_id
        item_name
        IFNull(Quantity, 0) As Quantity_checked
    from Inventory
```

Item_id	Item_name	Quantity Checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Question 12

Show employee ID, and the first available value among bonus or commission

Syntax

```
Select emp_id,
        Coalesce(bonus, commission) As first_available_reward
    from Employees_Extra;
```

Emp_id	First_available_reward
1	300
2	100
3	Null

Question 13

Count classes that dont have a room assigned

Syntax

```
Select Count(*) As no_room_count
    from Classes
    WHERE room is Null
```

NO_room_count
2

Q14

Show attendance records with Status. Replace Null with "Not Marked"

Syntax

```
Select Student_id,
        date,
    IfNull(Status,'Not Marked') As attendance_Status
From Attendance;
```

Student_ID	date	attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Question 15

Show account ID, account_type (or 'Unknown'), and balance (or 0)

Syntax

```
Select account_id,
        IfNull(account_type, 'Unknown') As type_display,
        IfNull(balance,0) As balance_Checked
From Bank_Accounts
```

Account_id	type_display	balance_Checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Question 16

Show all Projects with a Start date. If Start_date is Null, display 'TBD'

Syntax

```
Select Project_ID,
        title,
    IFNull (Start_date, 'TBD') As Start_display
  From Projects;
```

Project_id	title	Start_display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Question 17

Display reviews showing comment (or 'No Comment') and rating (or 0)

Syntax

```
Select review_id,
        Product_id,
    IFNull (Comment, 'NO Comment') As comment_display
    IFNull (rating, 0) As rating_display
  From Reviews;
```

review_id	Product_id	Comment_display	rating_display
1	501	Great Product	4
2	502	No Comment	0
3	503	Works fine	3

Question 18

Show the supplier contract number. Use Coalesce (Phone, alt_phone, 'No Contact')

Syntax

```
Select supplier_id,
        Name,
    Coalesce (Phone, alt_phone, 'No Contact') As Contact_number
From Suppliers;
```

supplier_id	name	Contact_number
1	Global Goods	123456789
2	Best Supplier	987654321
3	Value Source	No Contact

Question 19

Show all users and their prefferences. Replace all Nulls with defaults:

- theme - "light"
- Language - "English"
- Timezone - "Utc"

Syntax

```
Select user_id,
        IFNull (theme, 'light') As theme_set,
        IFNull (language, 'English') As language_set
        IFNull (timezone, 'UTC') As timezone_set
From user_settings;
```

User_id	theme_set	language_set	timezone_set
1	light	English	UTC
2	Dark	English	UTC+1
3	light	English	UTC

Question 20 : Show a maintenance log with:-

```
select record_id,
       machine_id,
   IFNull(issue, 'Unknown issue') As issue_log,
   IFNull(technician, 'Not Assigned') As technician_name
From maintenance;
```

record_ID	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown issue	Not Assigned
3	M103	Jammed	Alex